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AGR 101: Principles of Agriculture - Soils, Crops and Farming Systems

The science behind food production, from soil physics to farm planning

FACULTY PRIMER

Faculty Primer

Faculty of Agriculture
Crop Science & Horticulture | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

This pack introduces agriculture as applied science and business: soil formation and fertility, crop growth and management, animal production basics, farming systems and the realities of extension and markets in Nigeria.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Describe soil formation, texture, structure and fertility management.
- Explain the growth stages and management needs of a field crop.
- Compare farming systems and their suitability to Nigerian conditions.
- Identify the main constraints in Nigerian agricultural value chains.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

Soil as a living system

Soil is the medium of plant growth, and it is a biological system, not inert dirt.

- Soil formation involves weathering, organic matter accumulation and biological activity over time.
- Texture refers to the proportions of sand, silt and clay; structure refers to how particles aggregate.
- Porosity and water-holding capacity determine irrigation and drainage needs.
- Soil pH affects nutrient availability; most crops do best in a slightly acidic to neutral range.
- Organic matter improves structure, water retention and microbial activity.
- Fertility requires both nutrients and suitable physical and biological conditions.

> **Test before you apply**
> Applying fertiliser without a soil test is guesswork. Excess nitrogen can burn crops and pollute water, while a micronutrient deficiency may go untreated.

Crop production

Crop yield is the product of genetics, environment and management.

- Land preparation: clearing, tillage and seedbed formation appropriate to the crop.
- Propagation: seed selection, nursery practice, transplanting and spacing.
- Water management: irrigation scheduling based on crop need and soil moisture.
- Weed control: mechanical, cultural, biological and chemical methods, and integrated management.
- Pest and disease management: scouting, thresholds, resistant varieties and safe pesticide use.
- Post-harvest handling determines whether the yield reaches the market in saleable condition.

Systems, livestock and value chains

A farm is a system in which crops, animals, labour and money interact.

- Shifting cultivation and bush fallow rely on long fallow periods to restore fertility.
- Mixed cropping and intercropping spread risk and use space efficiently.
- Intensive systems require higher inputs and careful management.
- Livestock production depends on breeds adapted to local conditions, feeding, housing and health.
- Extension links research to farmers but is constrained by staffing and funding.
- Value chain constraints include storage losses, poor roads, credit scarcity and price volatility.

Agriculture as a system

A farm is a system with inputs (land, labour, capital, water, seed, fertiliser), processes (crop or animal production) and outputs (food, fibre, income, waste). Systems thinking reveals trade-offs: intensifying production may deplete soil; expanding area may require labour the household does not have.

Nigeria's agriculture is dominated by smallholders producing most of the food on small plots with limited mechanisation and rainfall dependence. Debates about productivity, credit, extension and market access are therefore debates about smallholder conditions.

The value chain runs from input supply through production, processing, storage, transport and marketing. Value is often captured downstream, which is why farmers who sell raw commodities earn least.

Soil, water and crops

Soil is a living system of minerals, organic matter, air, water and organisms. Texture determines water-holding capacity and drainage; structure determines root penetration and erosion resistance; pH determines nutrient availability. A soil test is the cheapest input decision a farmer can make.

Plants need light, water, carbon dioxide and sixteen essential nutrients. Nitrogen drives vegetative growth, phosphorus supports roots and flowering, potassium regulates water and quality. Deficiency symptoms overlap, which is why testing beats guessing.

Crop rotation, intercropping, cover cropping, mulching and organic matter management maintain fertility and interrupt pest cycles. These are agronomic principles that inputs should support, not replace.

> **Test before you apply**
> Fertiliser applied without a soil test is a guess with a price. The test costs less than one bag.

Livestock, pests and markets

Livestock production depends on feed, health, housing and genetics. Feed is usually the largest cost and feed conversion ratio is the key efficiency measure. Disease control requires vaccination, biosecurity and prompt reporting; zoonoses make animal health a public health matter.

Integrated pest management combines resistant varieties, cultural practices, biological control and judicious pesticide use. Indiscriminate spraying destroys natural enemies and selects for resistance.

Market access requires consistent quality, volume and timing. A farmer who produces well but cannot store, grade or aggregate with others sells at the lowest price of the season.

WORKED EXAMPLES

Do it once, slowly

These are not model answers to copy. They show the reasoning a strong student would use, step by step, so you can repeat the process on a question you have not seen.

1. Calculating a gross margin

Inputs for one hectare of maize cost 120,000; the harvest is 2.5 tonnes sold at 90,000 per tonne.

1. Revenue: 2.5 x 90,000 = 225,000.
2. Variable costs: 120,000.
3. Gross margin: 105,000 per hectare.
4. Check whether family labour and land rent are included.
5. Compare with an alternative crop on the same land.
6. Test a 20 per cent price fall: revenue 180,000, margin 60,000.

> **What to carry forward**
> Gross margin is not profit. State what is excluded before concluding.

2. Designing a rotation

A farmer grows maize continuously on three plots.

1. Identify problems: declining yield, pest build-up, nutrient depletion.
2. Choose a legume to fix nitrogen and break pest cycles.
3. Alternate cereal and legume across seasons on each plot.
4. Include a cover crop where land allows.
5. Plan residue management to protect soil cover.
6. State how success will be measured.

> **What to carry forward**
> Every rotation choice should name the problem it solves.

EXAM TECHNIQUE

Questions with model answers

Written in the shape your course is usually examined, with the marking logic spelled out. Attempt each one under time before you read the outline.

Q1. Explain the role of the three primary macronutrients in crop growth.

15 marks

1. Nitrogen: vegetative growth, chlorophyll, protein.
2. Phosphorus: roots, energy transfer, flowering.
3. Potassium: water regulation, enzyme activity, quality.
4. One deficiency symptom for each; note that excess is harmful.

Q2. Discuss the constraints facing smallholder farmers in Nigeria.

20 marks

1. Land tenure and fragmentation.
2. Credit and insurance access.
3. Input costs and counterfeit inputs.
4. Extension and information.
5. Post-harvest losses and market access.
6. One realistic intervention and its limitation.

REFERENCE

Glossary

Use these definitions in your own sentences; a copied definition rarely earns full marks.

Smallholder. A farmer cultivating a small area, often with family labour.

Value chain. Activities from input supply to final consumer.

Soil texture. The relative proportions of sand, silt and clay.

Soil pH. Acidity or alkalinity affecting nutrient availability.

Macronutrient. A nutrient needed in large amounts by plants.

Crop rotation. Growing different crops in sequence on the same land.

Intercropping. Growing two or more crops together on one field.

Integrated pest management. Combining methods to control pests with minimal chemical use.

Feed conversion ratio. Feed required to produce one unit of animal product.

Biosecurity. Measures preventing disease introduction and spread on a farm.

Gross margin. Revenue minus variable costs for an enterprise.

Post-harvest loss. Product lost between harvest and consumption.

Your four-week plan

Week 1 - Systems

The farm.

- Draw inputs, processes and outputs for a local farm.
- Identify two trade-offs.
- Map one value chain.

Week 2 - Soil

Fertility.

- Interpret a soil test report.
- Explain pH and nutrient availability.
- Design a rotation.

Week 3 - Production

Crops and animals.

- Identify three deficiency symptoms.
- Explain feed conversion.
- List biosecurity measures.

Week 4 - Economics

Decisions.

- Calculate a gross margin.
- Run a price sensitivity check.
- Write a policy answer.

Where to read more

Entry-level texts you can find in the university library or a departmental reading list. Confirm editions with your lecturer.

- Any principles of agriculture text covering soils, crops and livestock.
- A farm management guide on gross margins and budgeting.
- An extension manual on integrated pest management.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. What is soil texture?

Answer: The relative proportions of sand, silt and clay particles in a soil.

2. Why does organic matter matter?

Answer: It improves soil structure, water-holding capacity and biological activity, and supplies nutrients as it decomposes.

3. Give one advantage and one disadvantage of mixed cropping.

Answer: Advantage: risk spreading and efficient land use. Disadvantage: competition for light, water and nutrients and greater complexity in pest management.

4. What is integrated pest management?

Answer: A strategy combining cultural, biological, mechanical and chemical methods with monitoring and thresholds, using chemicals as a last resort.

5. Name two post-harvest constraints in Nigeria.

Answer: Inadequate storage leading to spoilage, and poor road infrastructure leading to delayed and damaged transport.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Describe the four components of soil.
- [ ] Test the texture of two soil samples by hand feel and record results.
- [ ] List the growth stages of a common Nigerian crop.
- [ ] Compare three farming systems in a table.
- [ ] Identify five pests or diseases of a staple crop.
- [ ] Draw a simple value chain for a crop you know.
- [ ] Interview a farmer about constraints and record five findings.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date:

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